HARSHIT AGARWAL

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EDUCATION

Jalpaiguri Government Engineering College, West Bengal Aug 2024 – Present
Bachelor of Technology in Electronics and Communication Engineering (ECE) *YGPA: 8.16*

EXPERIENCE

Indian Institute of Technology(IIT) Guwahati Jun 2026 – Present
Research Intern *Guwahati, Assam*

- Engineered high-throughput, end-to-end infrastructure for **NLP pipelines** (NER, POS Tagging, Text Classification) utilizing **Transformers** and **Tensorflow**, optimizing underlying data structures to improve tokenization efficiency.
- Architected scalable engineering solutions for NLP systems, executing performance-driven optimizations and debugging to ensure deterministic reproducibility and high-reliability production deployments.

PROJECTS

campAIgn: AI Marketing Campaign Automation Platform *React, Node.js, MongoDB, n8n, OpenAI API*

- Deployed a comprehensive email marketing solution, enhancing campaign creation and preview capabilities, resulting in over **500** successful campaigns executed through the platform.
- Designed high-performance **REST APIs** with authentication (**JWT**) ensuring secure synchronization of event-driven automation workflow triggers.
- Developed event-driven automation workflows using **n8n** to handle campaign generation, audience segmentation, and email delivery.
- Implemented customer segmentation pipeline using **Python (Scikit-Learn)** to process **5,000+** records and enable targeted messaging.

Distributed Edge Computing Optimization Simulator *Algorithm Design, Dataclasses, Object-Oriented Design*

- Engineered a highly scalable Mobile Edge Computing (MEC) simulation framework in **Python**, accurately modeling distributed task processing, network latency, and power constraints for high volume workloads of up to **100,000 concurrent tasks**
- Architected implementations of advanced metaheuristic algorithms including **Particle Swarm Optimization (PSO), Differential Evolution (DE)**, and a custom **Hybrid PSO+DE** to dynamically optimize task offloading policies
- Designed a rigorous, compute normalized benchmarking pipeline utilizing **NumPy** for vectorized state evaluation, analyzing and optimizing for critical system metrics such as **p99 tail latency**, energy efficiency, and server overload

Sentiment Classification from Text *Python, scikit-learn, NLTK, TF-IDF, Streamlit*

- Built a 6-class emotion classifier (joy, sadness, anger, fear, love, surprise) on **16,000+** sentences using **TF-IDF bigrams** and **Logistic Regression**, achieving 90% test accuracy.
- Benchmarked 3 approaches (**BoW+NaiveBayes, TF-IDF+NaiveBayes, TF-IDF+LR**) and selected the best-performing pipeline to handle class imbalance.
- Deployed an interactive **Streamlit** app with real-time emotion prediction and per-class confidence scores.

TECHNICAL SKILLS

- **Languages:** Java, Python, C
- **Core CS:** Data Structures and Algorithms (DSA), OOPs, OS, DBMS
- **Web Development:** HTML, CSS, JavaScript, Node.js, Express.js, EJS
- **Machine Learning & Deep Learning:** ANN, CNN, NLP, RNN (LSTM, GRU), Transformers (BERT, ELMO), LLM Fine Tuning, Token Classification, Computer Vision
- **Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Excel

RESEARCH & PUBLICATIONS

- **Metaheuristic & Cloud Synergy:** Co-authored 2 papers on task-offloading and differential evolution for MEC networks, accepted at **IEEE** and **Springer Nature**
- **Data Driven ML Applications:** Co-authored 2 additional accepted papers utilizing Apriori and tuned ML models for predictive forecasting (**Springer LNEE** / **ACSD 2025**)

ACHIEVEMENTS

- **Leetcode:** Solved **400+** DSA problems on LeetCode and various coding platforms
- Awarded a merit-cum-means scholarship by the Government of West Bengal
- Completed multiple online **Certifications** in relevant fields